CODECHELLA MADRID 2026

# A DiD Checklist

*Walking through the Brazil mental health reform, step by step*


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# Today’s example: the Brazil mental health reform

**Dias & Fontes (2024)**
*“The Effects of a Large-Scale Mental Health Reform: Evidence from Brazil”*
*American Economic Journal: Economic Policy*, 16(3): 257–289

**What they study.** Brazil’s 2002 psychiatric reform replaced large inpatient hospitals with *community mental health centers* (CAPS) — a major deinstitutionalization push.

**Why we use it here.** It is a clean staggered rollout, with a large panel of municipalities, and the right ingredients for a real DiD checklist.

# The reform in one slide

**What CAPS is.** A *Centro de Atenção Psicossocial* — a community mental health center providing outpatient care, day programs, and psychosocial support. The reform created CAPS centers as a substitute for inpatient psychiatric hospitals.

**Scale.** Over 1,600 CAPS centers opened across 5,180 municipalities between 2002 and 2016. A municipality is either treated (gets a CAPS in some year) or never treated.

**This is the canonical setup for staggered DiD:** many units, different treatment dates, a long enough panel to see pre- and post-trends.

# The headline finding

**What worked.**

- Increased **outpatient mental health care** use.
- **Reduced psychiatric hospitalizations**.

**What didn't.**

- ***Increased homicide rates*** — deinstitutionalization channel: vulnerable patients in the community without supervision.
- Hospital savings did not offset the social cost.

*We'll focus on the homicide-rate outcome.*

## A checklist for staggered DiD

*The Checklist Manifesto* (Gawande, 2009) makes the case that even highly skilled experts skip steps under cognitive load. Pilots, surgeons, structural engineers — all use written checklists. Empiricists should too.

**For staggered DiD, the steps are:**

1. Define the target parameter.
2. Tabulate when units are treated.
3. Plot the rollout.
4. Plot the outcome by cohort.
5. Select covariates — with discipline.
6. Inspect the propensity score.
7. Estimate (CS, BJS, . . .).
8. Run an event study (universal baseline).
9. Sensitivity analysis (Rambachan–Roth).

# Step 1: Define the target parameter

**What is the question?** Did opening a CAPS center, in the years after it opened, change the homicide rate in that municipality?

**Three ingredients** go into any target parameter:

1. A *difference in potential outcomes* $Y(1) - Y(0)$.
2. A *population* on which we average it.
3. A *weighting* rule $\mathbb{E}[\,\cdot\,]$.

*Most disagreements about "the" DiD estimand are disagreements about which of these three you chose.*

# The target: ATT on the treated, in post-treatment periods

$$\text{ATT} \;=\; \mathbb{E}\Big[\, \underbrace{Y_{it}(1) - Y_{it}(0)}_{\text{(1) PO contrast}} \;\Big|\; \underbrace{D_i = 1,\ \text{Post}_t}_{\text{(2) population}} \,\Big]$$

**In our case:**

- **Contrast.** Y(1) = homicide rate with a CAPS; Y(0) = homicide rate without one.
- **Population.** Municipalities that *actually got* a CAPS, after they got it.
- **Weighting.** Group-time ATTs $(g, t)$; we'll aggregate later.

**Decide this *first*.** Every downstream choice is meant to identify *this* parameter.

# Step 2: Tabulate when units are treated

Before anything else, count the cohorts: how many municipalities got a CAPS in each year, and how many were never treated.

**Why this table earns its place on the slide.**

- It defines the **groups** ($g$) in the estimator.
- It tells you which cohort is the *smallest* — which sets a binding constraint on covariate count (see Step 5).
- It tells you which units are *always treated* (must be dropped) and *never treated* (the control group).

*Next slide: the table itself.*

# CAPS introduction by cohort, 2003–2016

| Cohort | Municipalities | Share of treated | Max cov. (EPV $\leq$ 7) |
|---|---|---|---|
| 2003 | 47 | 3.5% | **6** (binds) |
| 2004 | 70 | 5.2% | 10 |
| 2005 | 95 | 7.1% | 13 |
| 2006 | **216** | **16.1%** | 30 (largest) |
| 2007 | 105 | 7.8% | 15 |
| 2008 | 112 | 8.3% | 16 |
| 2009 | 75 | 5.6% | 10 |
| 2010 | 96 | 7.1% | 13 |
| 2011 | 79 | 5.9% | 11 |
| 2012 | 114 | 8.5% | 16 |
| 2013 | 80 | 6.0% | 11 |
| 2014 | 97 | 7.2% | 13 |
| 2015 | 80 | 6.0% | 11 |
| 2016 | 78 | 5.8% | 11 |
| **Treated total** | **1,344** | **100%** | — |
| Always-treated (2002, excluded) | 296 | — | — |
| Never-treated controls | 3,836 | — | — |

*2003 binds covariate count; 2006 dominates the simple-aggregation weight.*

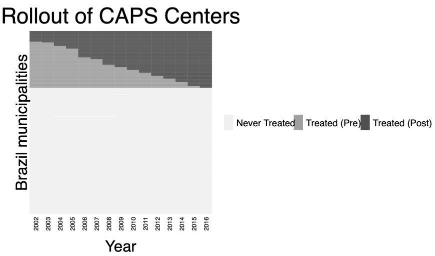
## Step 3: Plot the rollout

Before any estimator, look at *when* treatment turns on, unit by unit.

**Why look at it.** The rollout picture tells you whether the staggering is rich enough to support staggered DiD (some units treated, some not, treatment dates spread across years), or whether you have a near-event-study problem (most units treated in one or two years).

*Tool: `panelview` (Yiqing Xu, Stanford). Rows are municipalities, columns are years, color encodes pre vs post treatment.*

# Brazil CAPS rollout: 5,476 municipalities, 2002–2016




*Light gray = never-treated. Medium gray = treated, pre-period. Dark gray = treated, post-period.*

## Step 4: Plot the outcome by cohort

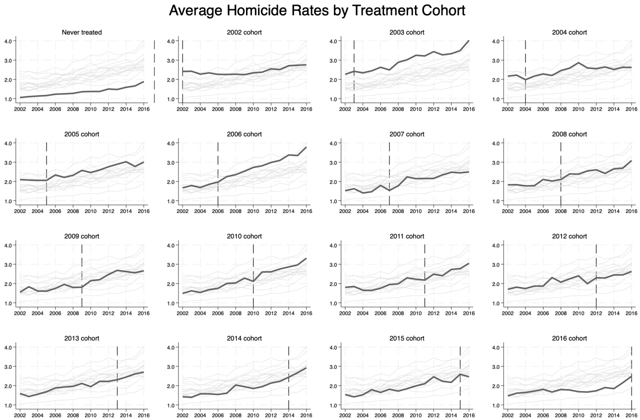
Now plot the *homicide rate* over time, separately for each cohort. Two purposes:

- See whether trends move in parallel across cohorts in the pre-period.
- Catch any anomaly — a missing year, a level break, an outlier cohort — before the estimator hides it from you.

**One important reminder.** DiD identifies $Y(0)$'s *trend*, not its *level*. Levels are differenced away. You are looking for parallel *movement*, not for cohorts that overlay each other.

*This is the difference between DiD and synthetic control. Synth requires comparable levels and trends; DiD only needs comparable trends.*

# Homicide rate by treatment cohort




*One small-multiples panel per cohort. Bold dark line = focal cohort. Dashed vertical = treatment year.*

## Step 5: Two ways to think about covariates

**Path 1 — Outcome Regression (OR).** Pick covariates that predict $Y(0)$'s *trend*. Ask: what are the ordinary drivers of homicide trends in Brazilian municipalities? Income, demographics, poverty, urbanization, federal transfers, education.

**Path 2 — Propensity Score (PS).** Pick covariates that predict *treatment timing*. Ask: who got a CAPS first? Larger cities? Poorer states? Earlier hospital capacity?

**Doubly robust uses both.** If you can correctly specify either the outcome trend or the propensity score, DR is consistent. You don't have to be right twice.

*Use common sense first. Run LASSO or stepwise selection only after a domain-informed shortlist.*

# Step 5 (cont.): the 7-events-per-covariate rule

**Peduzzi, Concato, Kemper, Holford & Feinstein (1996),** *"A simulation study of the number of events per variable in logistic regression analysis," Journal of Clinical Epidemiology*, 49(12): 1373–1379.

**The rule.** A logistic regression needs roughly 7–10 "events" (the smaller of the two outcome classes) per covariate to avoid overfit and unstable coefficients.

**In a propensity score model for staggered DiD,** the "event" is the number of treated units in a 2$\times$2 comparison. If cohort $g$ has 47 municipalities (our 2003 cohort), the EPV ceiling is roughly $\lfloor 47/7 \rfloor = 6$ covariates.

**Different 2$\times$2's can carry different covariate counts. The smallest cohort sets the binding limit.**

## Step 5 (cont.): standardize, for the computer's sake

**The problem.** Some covariates look like `poptotaltrend = population` $\times$ `year`. In Brazil that can be 150 million or larger. When the propensity-score logit tries to invert a matrix with entries that big, it doesn't fail like a textbook tells you — it fails like a computer.

**The fix.** Convert every continuous covariate to a *z-score*: subtract the mean, divide by the SD. Now everything is on the same scale, the matrix inverts cleanly, and your estimates don't shift just because someone changed units.

**This is not econometrics. It is numerical hygiene.** The model is the same; only the computer's life is easier.

# Step 5 (cont.): the balance table the paper actually used

The published spec used **31 covariates** in the propensity score and outcome model. The balance table (baseline year 2002, excluding the always-treated 2002 cohort) reports a normalized difference for each one.

### 31 candidates included things like

- Log GDP per capita
- Federal cash-transfer receipts
- 14 age $\times$ sex population bins
- Rural population share
- Pre-period homicide trend
- Pre-period health-spending trend

### Imbalance is real.

- 9 covariates with $|$norm. diff.$|$ $>$ 0.25
- Largest: `poptotaltrend` at $+0.69$
- Smallest cohort (2003): EPV cap of 6
- Largest cohort (2006): EPV cap of 30

**31 candidates, 6 slots in the binding 2$\times$2.** Step 5 picks which 6.

## Step 6: Inspect the propensity score

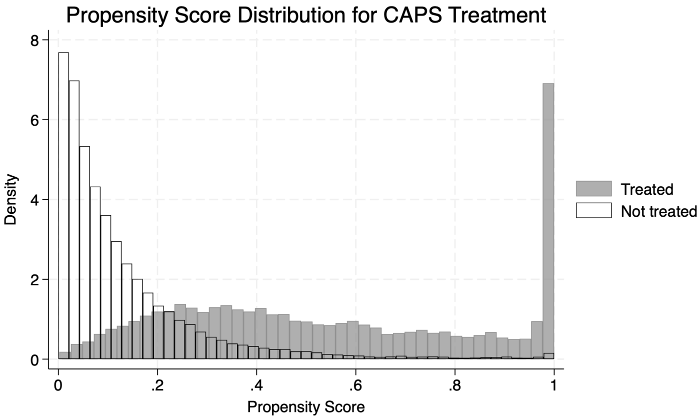
After picking covariates, fit `probit treat $controls if ano==2002` and plot $\widehat{p}(X)$ for treated and untreated municipalities.

**What you want to see:** substantial **overlap**. Treated and untreated propensity-score distributions should share most of their support. Mass at $\widehat{p} \approx 1$ for controls is a red flag — those are units the model thinks should have been treated but weren't.

**Standard practice:** trim controls with $\widehat{p} > 0.995$ before estimating. The `did` R package does this automatically.

*Next slide: histogram from a fitted PS in our setup.*

# Propensity score distribution, by treatment status




*Gray = treated. Black outline = never-treated. Mass in the right tail of controls is the trimming target.*

## Step 7: Estimate with the Callaway–Sant'Anna estimator

Three choices, made deliberately:

- **Baseline period.** Use `long2` in Stata / `base_period = "universal"` in R. This pins every event-time effect to $t = -1$, so pre-trend coefficients are real falsification tests.
- **Estimation method.** `dr` (doubly robust) or `ipw` or `reg`. DR if you trust neither model individually; OR if you don't want to extrapolate from a logistic regression with overlap problems.
- **Control group.** `notyet` — not-yet-treated comparison. Bigger sample than never-treated only; no pre-trend restriction.

# Step 7 (cont.): the aggregation question

`csdid` returns group-time ATTs: $\widehat{\text{ATT}}(g, t)$ for every cohort $g$ and post-period $t$. Aggregating these is a choice with consequences.

- **Simple ATT.** Average over all $(g, t)$ post-treatment cells, weighted by cohort size $\times$ exposure length.
- **Group ATT.** First average within cohort to one number per $g$; then average across cohorts.
- **Event-study ATT.** Average across cohorts at each event time $e = t - g$.

**Big cohorts have big weights.** The 2006 cohort (216 units, 16% of treated) will dominate any size-weighted average.

# Step 7 (cont.): a small artifact, to make weighting concrete

Two cohorts. **Group A**: 5 post-period cells, 500 units each. **Group B**: 2 post-period cells, 100 units each.

| | Group A (n = 500) | | | | | Group B (n = 100) | |
|---|---|---|---|---|---|---|---|
| cell | $A_1$ | $A_2$ | $A_3$ | $A_4$ | $A_5$ | $B_1$ | $B_2$ |
| $\widehat{ATT}$ | 0.10 | 0.12 | 0.14 | 0.11 | 0.13 | 0.50 | 0.60 |

**Both aggregations weight by population — just at different units:**

- **Simple** (each *cell* weighted by $N_g$): $\frac{5 \cdot 500 \cdot 0.12 + 2 \cdot 100 \cdot 0.55}{5(500)+2(100)} = \frac{300+110}{2700} =$ **0.152**
- **Group** (each *cohort* weighted by $N_g$): $\frac{500 \cdot 0.12 + 100 \cdot 0.55}{600} = \frac{60+55}{600} =$ **0.192**

**Simple amplifies long-exposure cohorts; group neutralizes exposure length.**

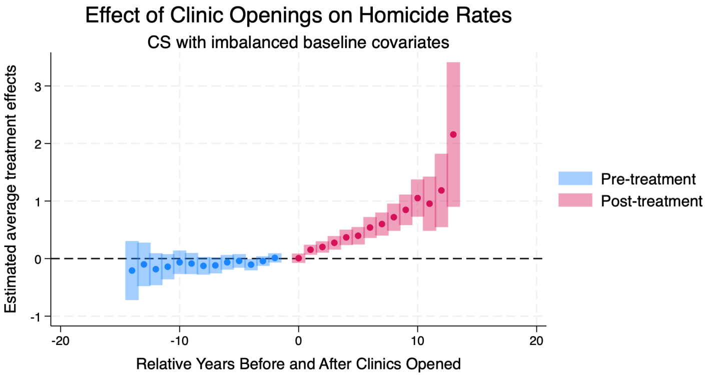
# Step 8: The event study, with a universal baseline

The event study reports $\widehat{\text{ATT}}(e)$ for each event time $e = t - g$. Two reasons we use a *universal* baseline ($t = -1$ pinned to zero):

- **Pre-trend coefficients become falsification tests.** Each pre-period $e < 0$ is now a "long difference" relative to $t = -1$. The identifying assumption is that *post-period* long differences in $Y(0)$ are zero; pre-period ones should be too.
- **Post-period coefficients become interpretable on the same scale.** Every $\widehat{\text{ATT}}(e)$ for $e \geq 0$ is the effect of an additional $e + 1$ years of treatment exposure relative to the year before.

*In Stata:* `csdid ... long2`. *In R:* `base_period = "universal"` *inside* `att_gt()`.

# The event study: CAPS effect on homicide rates




Note: Uses the 2008–2010 eventually treated and the never treated counties as controls. Circles are ATT(l) estimates by relative time, l. All groups and bands are 95% uniform confidence intervals. Mean homicide rate was approximately 1.23 in 2002.

*Years relative to CAPS opening. Bars are 95% uniform confidence intervals. Pre-trend points near zero pass the falsification test.*

## Step 9: Sensitivity — when pre-trends are imperfect

Pre-trends rarely look *exactly* flat. The question is then: how large would the parallel-trends violation have to be for our conclusion to flip?

**Rambachan & Roth (2023),** *"A More Credible Approach to Parallel Trends,"* *Review of Economic Studies*, 90(5): 2555–2591.

**The idea.** Treat the magnitude of post-treatment parallel-trends violation as bounded by some multiple $M$ of the largest pre-period violation. For each $M$, compute the *robust* confidence interval for the post-treatment effect. Then plot the CI as $M$ grows.

**The estimator is no longer point-identified — it's a set.** Honest about what the data can and cannot rule out.

## Step 9 (cont.): the code

The Stata implementation is the `honestdid` package (Rambachan, Roth, Wing). Two steps:

```
* 1. Estimate the CS event study with long2 / universal baseline
csdid2 homicide_rate $controls, gvar(g) ivar(cod) ///
    time(ano) long2 method(drimp) notyet
estat event, window(-4 5)

* 2. Run the sensitivity analysis on the post-period average
matrix l_vec = J(1, 1, 1) // weight on Post_avg
honestdid, pre(3/5) post(2/2) ///
    mvec(0(0.25)2) delta(rm) ///
    l_vec(l_vec) b(e(b)) vcov(e(V))
```

- `mvec(0(0.25)2)`: sweep $M$ from 0 to 2 in steps of 0.25.
- `delta(rm)`: relative-magnitudes restriction (the standard choice).

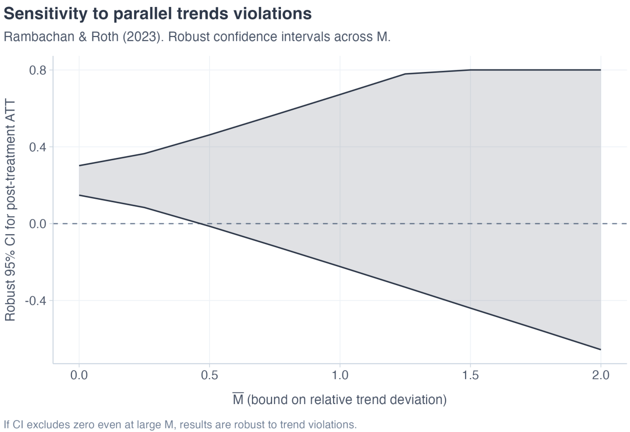
# Reading the HonestDID plot

The output is a forest of confidence intervals, one for each $M$.

- $M = 0$: the original CS estimate.
- $M = 0.5$: PT violation half as large as worst pre-period.
- $M = 1$: as large as worst pre-period.
- $M = 2$: twice as large.

**Breakdown point:** the $M$ at which the CI just touches zero. Surviving $M \geq 1$ means robust to anything seen in the pre-period.

# The HonestDID plot for CAPS




*Each bar is a 95% robust CI for the post-treatment ATT at a value of $M$. Black diamond at $M = 0$ is the original CS estimate.*

## A counterpoint: "Don't Do DiD"

**Guido Imbens** (Stanford, Nobel 2021) recently gave a talk titled *"Don't Do Difference-in-Differences"* — DDDiD.

**His argument.** Conditional parallel trends is a strong, untestable assumption. If you don't believe it, an estimator that requires it is the wrong tool. He recommends other panel methods — chief among them **synthetic control** and **synthetic DiD**.

*Imbens,* `youtube.com/watch?v=GG627T4GbqU`

# Why he might be right — and the 1970s caution

**The empirical crisis of the 1970s** was not a failure of econometrics. The math of 2SLS, regression, and instrumental variables is fine. Smart people could prove these results standing on their heads.

**What failed was the marriage of method to data.** Orley Ashenfelter and Janet Currie recall a time when researchers wouldn't even *disclose what their instrument was*. Identification was assumed, not defended.

**The more complex the estimator, the higher the returns to knowing your data.** And the higher the cost of being wrong about its assumptions.

## But beware absolutes

Many people will tell you that **including covariates in a DiD is always suspect.** Or that **you should never use TWFE under staggered adoption.** Or that **conditional parallel trends is too strong to defend.**

**Counter-example.** In the LaLonde–Dehejia–Wahba data, we just saw that conditional-on-$X$ DiD with the right covariates *does* recover an estimate close to the RCT benchmark. Proof by contradiction.

**Don't throw rocks in glass houses.** The smart move is to know your data well enough to know when conditional PT might hold, when it might not, and to argue the case in both directions.

# Tomorrow: when you don't believe conditional PT

If you have reason to doubt conditional parallel trends — and you can't shore it up with covariates, sensitivity analysis, or design — then you should be looking at a *different* class of estimator.

**Tomorrow we cover:**

- **Synthetic control** (Abadie, Diamond & Hainmueller).
- **Synthetic DiD** (Arkhangelsky, Athey, Hirshberg, Imbens & Wager).

**These trade one set of assumptions for another — and that trade is the conversation worth having.**